

Experiments continued

What matters for compatibility recovery?

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Agenda

Topics for today

Recap

Results

- Source-only learned utility family
- Source-only AE family
- Target-local direct support-NELBO

Key takeaways

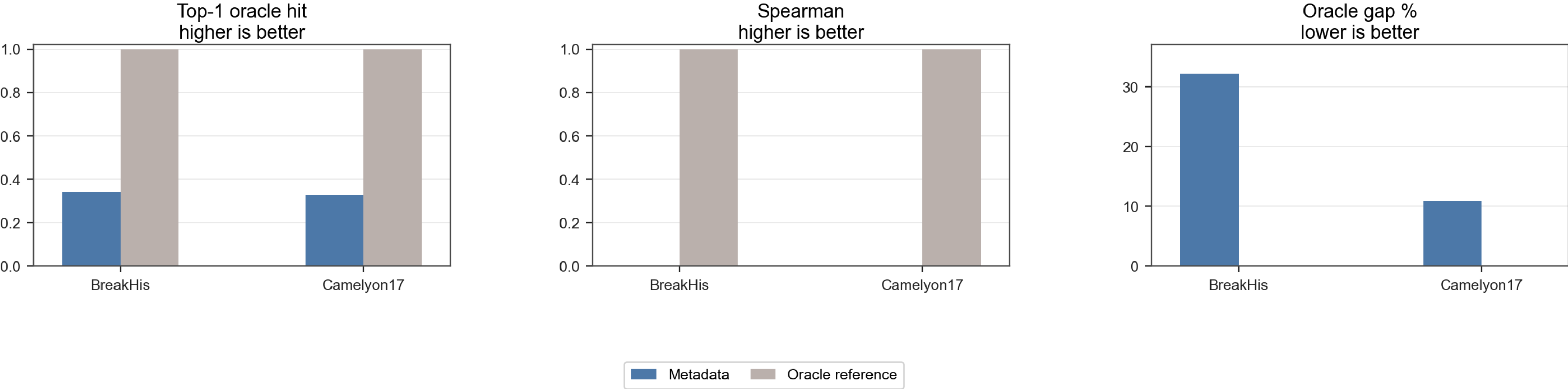
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Recap

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Metadata-Baseline

Metadata similarity does not suffice for compatibility routing



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Where we stopped last time

Key points

- Metadata routing changes expert selection but does not reliably improve utility
- Learned proxy models and metadata-conditioned variants did not recover utility-aligned rankings
- The bottleneck was localized to compatibility estimation, not the routing mechanism itself

The new question

How can we estimate compatibility without leakage?

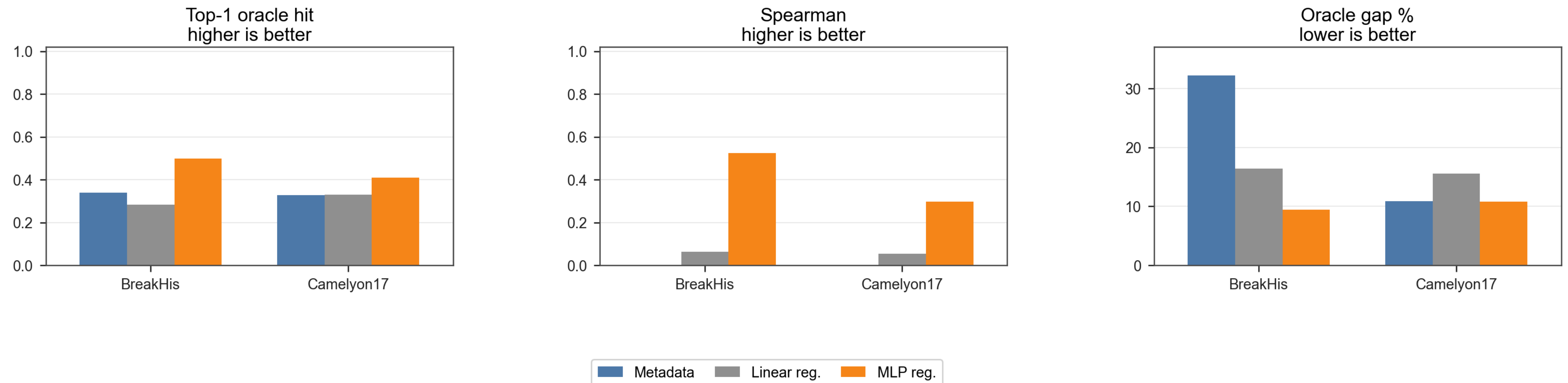
- Learned utility predictors
 - Pairwise ranker
- Source-only AE confidence family
 - AE utility calibrator
- Target-local direct support-NELBO

Results

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Result A

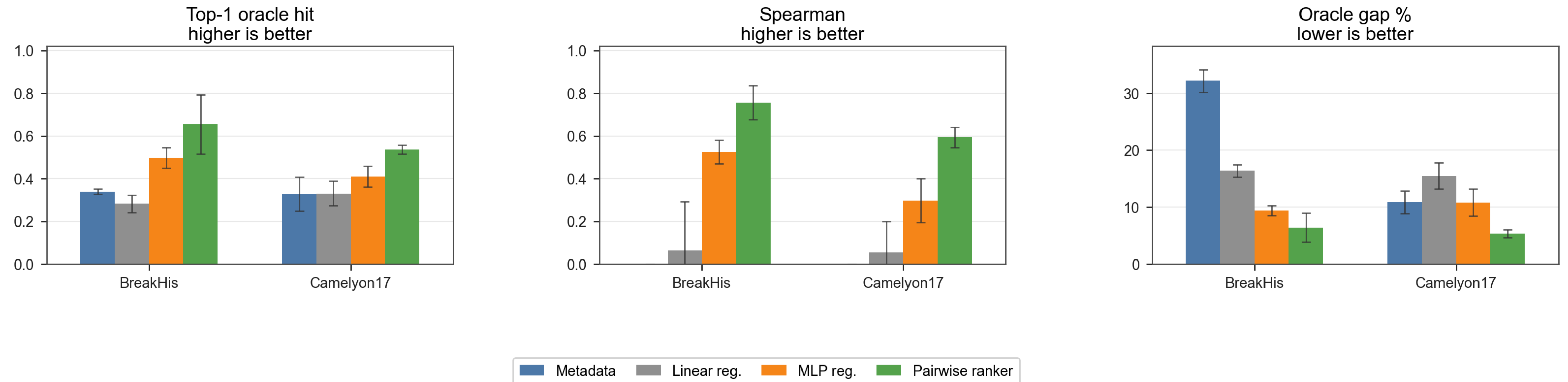
Learned predictors recover signal, but fail adoption gates



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Result A

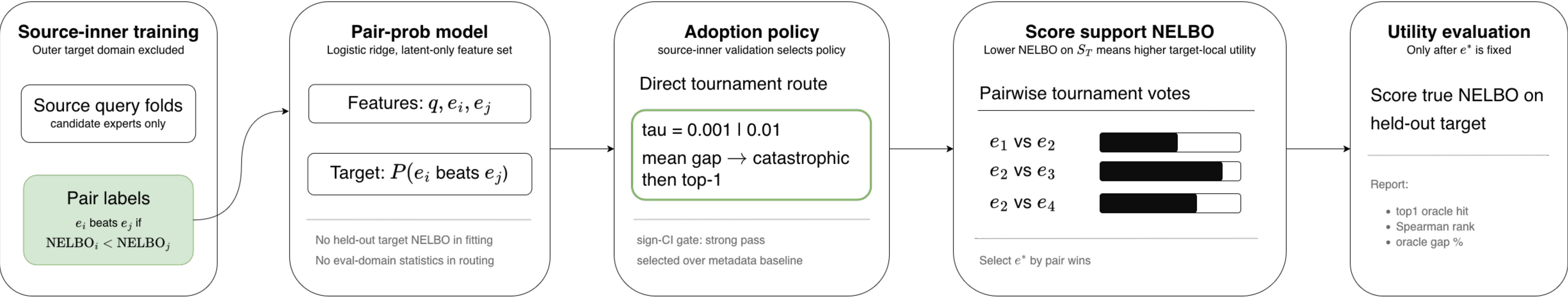
Learned predictors recover signal, but fail adoption gates



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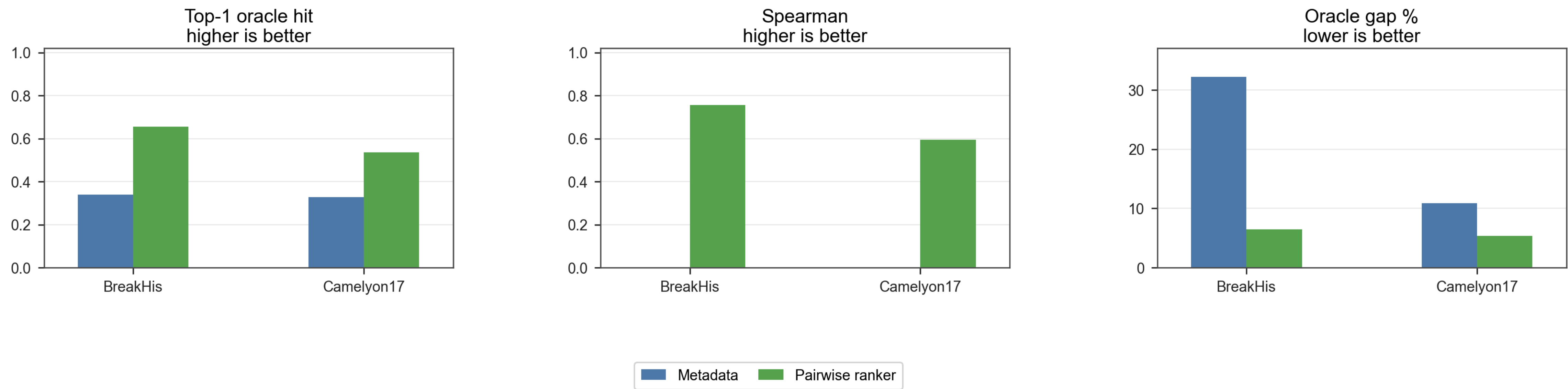
Pairwise probability tournament routing

Estimate who beats whom and select the tournament winner



Result B

Pairwise tournament improves ranking, but remains unstable

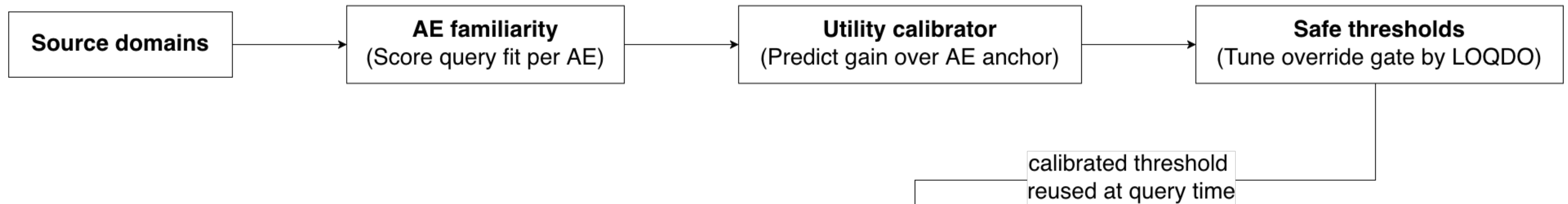


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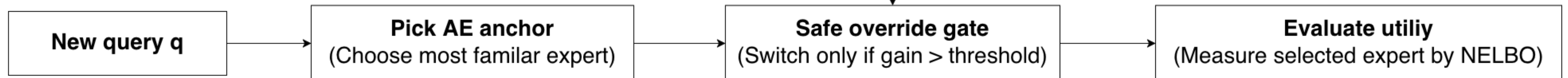
AE Utility calibrated safe override

Source-only AE familiarity becomes a gated routing decision

1. Source-only calibration loop



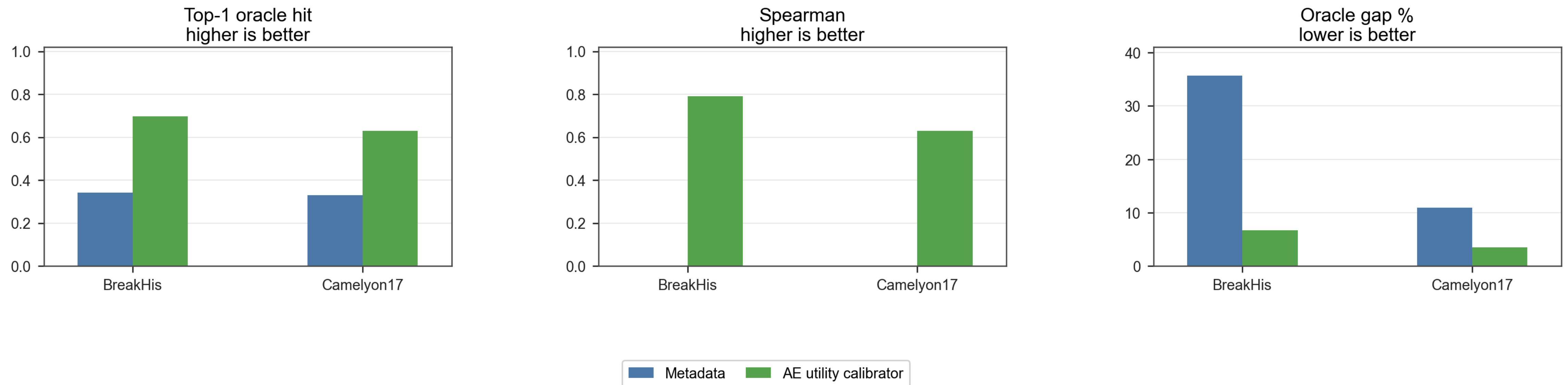
2. Query-time routing and evaluation



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Result C

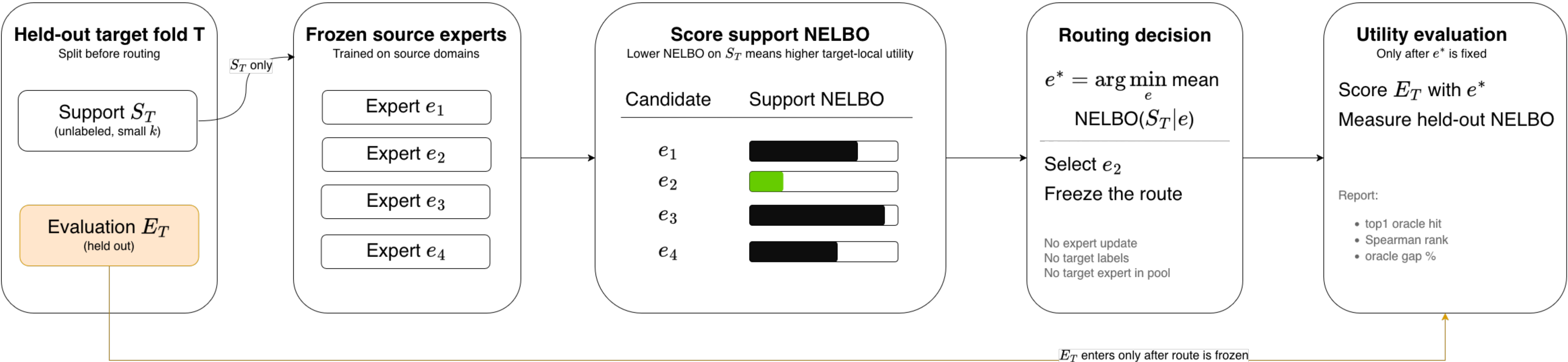
AE calibration helps, but remains dataset-dependent



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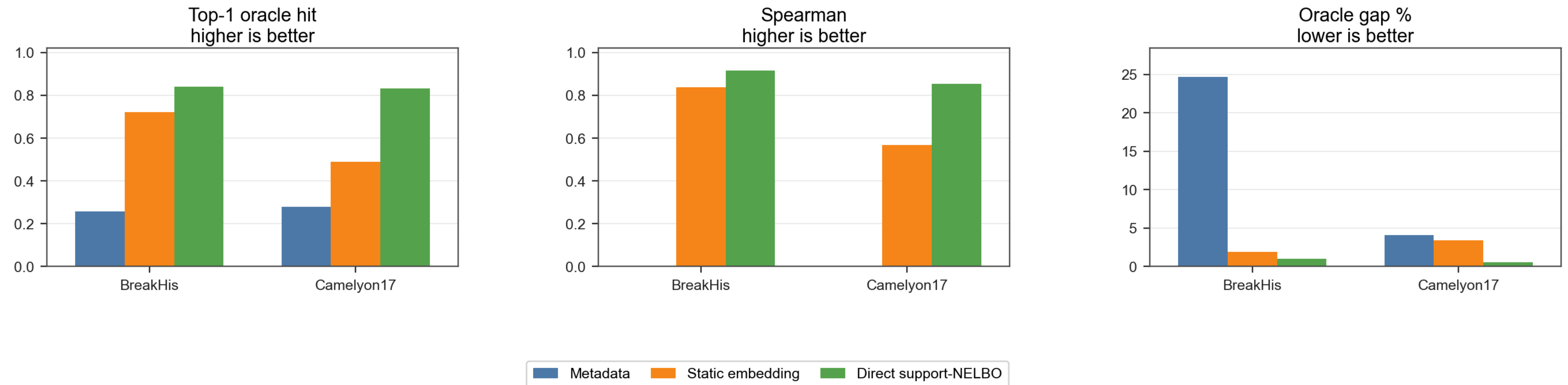
Direct support-NELBO routing

Pipeline regarding the target-local routing



Result D

Target-local NELBO turns compatibility into measurable utility



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Key takeaways

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Key takeaways

Hypothesis	Methods	Outcome
Domain identity is enough	Metadata routing	No
Source-only utility proxies are enough	Learned utility, pairwise tournament, AE confidence, AE calibrator	Useful but unstable
Target-local utility is needed	Direct support-NELBO	Strongest thesis-facing result

Thank you!